

TOWSON UNIVERSITY
COLLEGE OF GRADUATE STUDIES

AGENTIC ENVIRONMENT SYNTHESIS: EVALUATING THE EFFICACY OF
MODEL CONTEXT PROTOCOL (MCP) IN REDUCING HALLUCINATIONS OF
DECLARATIVE SYSTEM CONFIGURATION.

by

(Luis Dale Gascon)

A dissertation

Presented to the faculty

of Towson University

in partial fulfillment

of the requirements for the degree

Doctor of Philosophy

Department of Computer and Information Sciences

Towson University,

Towson, Maryland

21252

(Month Year of Graduation)

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Abstract

AGENTIC ENVIRONMENT SYNTHESIS: EVALUATING THE EFFICACY OF MODEL CONTEXT PROTOCOL (MCP) IN REDUCING HALLUCINATIONS OF DECLARATIVE SYSTEM CONFIGURATION.

(Luis Dale Gascon)

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Keys to Symbols, Abbreviations, or Nomenclature

[Optional - If used, the layout is up to the discretion of the student in consultation with the thesis/dissertation chairperson. The symbols, abbreviations, and nomenclature are to be consistent with those specified by the academic discipline and style manual.]

Chapter I.

How To Use This Template

Read the inline comments in the latex project for notes and explanations.

1.1 Required Pages

Preliminary and supplementary pages are labeled in the order they should appear. Optional pages are marked as optional in the comments. List of Tables and List of Figures are required only when 2 or more tables/figures appear in the document.

1.2 Paragraph Styles

In latex/overleaf, use `\chapter` for chapter headings, use `\section` for level 2 headings, use `\subsection` for level 3 headings, use `\subsubsection` for level 4 headings. Read the inline comments for detailed commands. Do not use other commands for headings!

1.3 Table of Contents, List of Tables, and List of Figures

A Table of Contents, List of Tables, and List of Figures are set up already. See Dissertation.tex and the inline comments for details!

1.4 How to use references

In latex/overleaf, use `\cite{}` for reference usage and put your bib text in the reference folder. Here is an example [1].

And you can also use `\ref{}` to refer to a chapter/section/subsection/subsubsection or figures/tables with the label. Here is an example: this is Chapter I.

Chapter II.

IEEE Style Headings

This chapter uses IEEE style headings with numbering. You will use the same heading styles with or without numbering, but this chapter is here to show you what the numbering should look like. Read the inline comments in the latex project for how to switch between this two.

2.1 IEEE Style Heading Level 2

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2.1.1 IEEE Style Heading Level 3: Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aliquam ac malesuada leo. Aliquam sit amet eros eleifend, molestie quam eu, scelerisque tortor. Etiam vel diam rutrum, maximus arcu nec, tempus magna. Praesent pretium, mauris eget euismod facilisis, libero lorem laoreet leo, at bibendum arcu massa at dui. Aliquam id felis mi. Vivamus egestas efficitur metus, sed semper libero facilisis in. Cras id scelerisque quam, sit amet dictum tellus.

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Chapter III.

Figures

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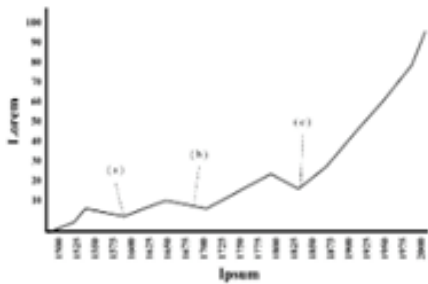


Fig. 1. Tempor nec feugiat nisl pretium fusce id velit. (a) Sed egestas egestas fringilla phasellus. (b) Turpis tincidunt id aliquet risus feugiat in. (c) Dolor morbi non arcu risus quis varius.

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Fig. 1. IEEE Style Figure Caption.

This is an example of how to include a figure in your writing. And you can also refer to this figure as Fig. 1. Read the inline comments.

Chapter IV.

Tables

Table I
IEEE Style Table Heading

Column 1	Column 2	Column 3
Abcde	123	1234
Fghijk	324	216*
Lmnop	325	0
Qrstuv	9578	33

Below the table, you may include explanatory notes as necessary.

* Notes that apply to a specific part of the table should be marked in the same manner as a footnote.

This is an example of how to include a table in your writing. And you can also refer to this table as Table I. Read the inline comments.

Chapter V.

Landscape Pages

Some tables and figures may fit better on a landscape-oriented page. When using a landscape page, the page numbers must be rotated so that they will appear in the correct orientation when the document is printed and bound. The following page is an example of a correctly numbered landscape page.

(Landscape Page Example)

This content is in landscape mode, and the page number is rotated as well. Read the inline comments for details of how to enable the landscape mode.

Chapter VI.

Examples of Some Special Cases

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This is one of the special cases, we have included the Fig. 2 before this paragraph in the code editor. However, the space in the rest of the page is not enough. In this case, the figure will be arranged the first available place in the next page, even interrupting the paragraph afterwards. To solve this, the best way is to manually change the place where the figure is included in the code editor.

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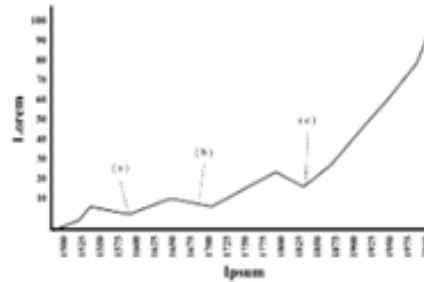


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Fig. 2. Figure Interrupting The Paragraphs

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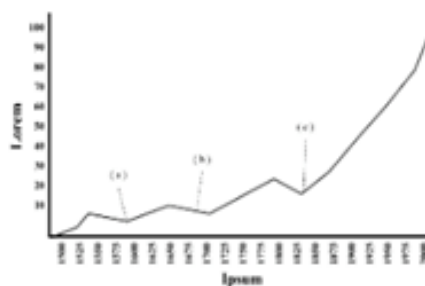


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Fig. 3. Figure Not Interrupting The Paragraphs

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Another special case is, when writing some formula or some explanations, you may write in this way: Lorem ipsum dolor sit amet, consectetur adipiscing adipis elit. here we denote the second as s .

You will find that the character s is isolated at the beginning of a new line. This should be avoid. The way to solve this is also included in the very beginning of Dissertation.tex. You can find the solution using `~` in the code editor in the next paragraph.

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Another special case is, when writing some formula or some explanations, you may write in this way: Lorem ipsum dolor sit amet, consectetur adipiscing adipis elit. here we denote the second as s .

In this way writing as “`as~s`” in the code editor, “`as s`” will be considered to be rendered together to avoid the isolated character in a new line. It is also recommended always connect your references or refers with a `~`. Read the code editor for details, e.g. Lorem ipsum [1], Lorem ipsum Fig. 3.

Appendix A

Appendix Title Here

[Appendices follow the last page of the text. Each appendix should be labeled, either at the top or on a proceeding blank page, as Appendix A, Appendix B, etc. The sequence of the appendices follows the order in which they were first introduced in the main body of the text. Each appendix needs to be labeled and named in the main body of the text for it to be included in the appendix section.]

Appendix B

Appendix Title Here

[Appendices follow the last page of the text. Each appendix should be labeled, either at the top or on a proceeding blank page, as Appendix A, Appendix B, etc. The sequence of the appendices follows the order in which they were first introduced in the main body of the text. Each appendix needs to be labeled and named in the main body of the text for it to be included in the appendix section.]

References

- [1] W. Y. (Editor and Co-Authors), Edge Intelligence in Cyber-Physical Systems: Foundations and Applications. Academic Pressr, June 1, 2025.

